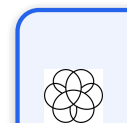
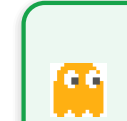


Motivation

- Educational tool that allows users to learn Deep Reinforcement Learning (DRL) concepts through hands-on exploration
- Built-in framework allows DRL agents to train on Atari-style games
- Training process and results are abstracted into an easy-to-use dashboard

Technologies

 **Gymnasium** (Atari API, v0.29.1)

 **ALE** (Arcade Learning Env, v0.8.1)

 **Stable-Baselines3** (v2.3.0)

 **TensorBoard** (≥ 2.14)

 **React** (v19)

 **Next.js** (v16)

 **Tailwind CSS**

 **TypeScript**

Team

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Milwaukee School of Engineering | 2025–26
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Application Flow

Entry point

Overview of Deep RL concepts and quick links to start a run.

Environment

Game registered via Gymnasium / ALE; frames stream at ~ 30 Hz.

Agent choice

Value-based (DQN, Rainbow) or policy-gradient (PPO). Each has trade-offs in stability and sample efficiency.

Tuning

Controls exploration vs. exploitation, credit assignment, and replay. Sensible defaults per algorithm.

Live comparison

REST drives the human session; a WebSocket streams the agent's training frames in parallel.

Analysis

Sortable tables of episodes and metrics by game and agent, side-by-side with the user's own score.

Home Page

Welcome screen with project overview; users press **Start** to enter the flow.

Select page

Browse and search Atari games, preview art, and persist the chosen environment for the run.

Algorithm page

Pick DQN, Rainbow, or PPO; each card summarizes the family (value-based vs. policy gradient).

Configure page

Tune hyperparameters (learning rate, γ , replay buffer, ...) filtered by the chosen algorithm.

Play page

Human plays via REST stepping/rendering; optional side-by-side WebSocket stream for a live agent preview.

Results page

Explore aggregate metrics, sortable tables, and per-episode details to compare games and agents.

UI Walkthrough

